Stock recruitment relationships for South African anchovy

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The assessment of the South African anchovy resource has recently been updated to inform the Operating Model of the upcoming joint sardine-anchovy Management Strategy Evaluation (de Moor 2025). This document considers alternative stock recruitment relationships estimated during the conditioning of the anchovy assessment, particularly in the light of three consecutive years of poor anchovy recruitment in the most recent years – the most recent of which (June 2025) is not included in the data to which this model is fit. When different stock recruitment relationships are fit to different time periods, the relationship corresponding to the most recent time period (2012-) is less productive than that estimated for the full time period.

Keywords: anchovy, assessment, stock recruitment relationships, South Africa

Introduction

The integrated assessment for South African anchovy is age structured, fitting to length structured data and detailed in de Moor (2025a). It is conditioned on survey and commercial data from 1984 to 2024 (de Moor *et al*. (2024), de Moor 2025).

While the Small Pelagic Scientific Working Group has previously favoured using a time-invariant stock recruitment relationship for a baseline stock assessment and baseline operating model for joint sardine-anchovy Management Strategy Evaluations, the most recent years of consecutive poor recruitment (including as surveyed in June 2025, but not included in this model) have resulted in reconsideration of this baseline assumption.

The assessment has been conditioned assuming the following stock recruitment relationships:

A_{BH} - Time-invariant Beverton Holt stock-recruitment relationship.

A_{2BH} - Two Beverton Holt stock-recruitment curves, one estimated from 1984 to 1998 and 2013 to 2024, and the other from 1999 to 2012.

A_{3BH} - Three Beverton Holt stock-recruitment curves, estimated from 1984 to 1998, 1999 to 2012 and 2013 to 2024. (This is included more as a means of understanding than a viable option.)

A_{HS} - Time-invariant Hockey-Stick stock-recruitment relationship.

A_{2HS} - Two Hockey-Stock stock-recruitment curves, one estimated from 1984 to 1998 and 2013 to 2024, and the other from 1999 to 2012.

Results and Discussion

The estimated Beverton-Holt stock recruitment relationships are shown in Figure 1 and the corresponding timeseries of October recruitments are shown in Figure 2 (noting recruitment in the final year may not be well estimated given the data available; recruitment in October 2023 for A_{BH} is substantially less than that estimated by de Moor (2024)).

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While there appears to be no trend in residuals against spawner biomass for A_{BH}, there is a decreasing trend in recruitment residuals in recent years (Figure 1a). This is not completely resolved for A_{2BH} (Figure 1b), but there is little remaining trend for A_{3BH} (Figure 1c). When two Beverton-Holt stock recruitment relationships are assumed (A_{2BH}), the fit to the November survey data is slightly worse, particularly in 2000-2002, 2011, 2024 (Table 1, de Moor 2025), but this is offset by an improved fit to the other data and a substantial reduction in the recruitment residuals (Table 1, de Moor 2025). The curve estimated for the early and later years is substantially less productive than that estimated by A_{BH} (Figure 1b), although the model simply estimated a flat median relationship for the middle years.

The model struggles to estimate three separate Beverton Holt relationships (Figure 1c).

The Beverton Holt stock recruitment relationship has been used for anchovy since 2012 after a considered change from the Hockey Stick stock recruitment relationship, particularly given the risk of no change to median biomass while spawner biomass decreases when assuming the Hockey Stick stock recruitment relationship with a relatively low hinge point (e.g. de Moor and Butterworth 2012). If switching back to a time-invariant Hockey Stick stock recruitment relationship, the hinge point is now estimated to be relatively high for this deterministic model fit. (Note that some local minima were found for the Hockey Stick stock recruitment relationship which wasn't evident when estimating the Beverton Holt relationship.) The comparison between A_{HS} and A_{HS2} is similar to that between A_{BH} and A_{BH2} in terms of model fit and a substantial difference in relationships estimated for the early/later periods compared to the middle period. The residuals for A_{HS} and A_{2HS} also show a similar pattern as A_{BH} and A_{2BH}.

The choice of stock-recruitment relationship to use for projections could be key given the relationships here corresponding to the most recent years might have a substantial impact on projections with the Management Strategy Evaluation.

References

de Moor CL. 2025. Assessment of the South African anchovy resource using data up to 2024. International Stock Assessment Workshop Document IWS/2025/Sardine/2025/BG7. (Also FISHERIES/2025/OCT/SWG-PEL/55.)

de Moor CL. 2024. Initial results towards a new 'benchmark' assessment of the South African anchovy resource. DFFE: Branch Fisheries Document FISHERIES/2025/AUG/SWG-PEL/35.

de Moor CL and Butterworth DS. 2012. Assessment of the South African anchovy resource using data from 1984-2011: results at the posterior mode. DAFF: Branch Fisheries Document FISHERIES/2012/SWG-PEL/42.

de Moor CL, Merkle D and Coetzee J. 2024. The data used in the 2024 anchovy assessment. DFFE: Branch Fisheries Document FISHERIES/2024/AUG/SWG-PEL/34.

Table 1. The contributions to the negative log posterior pdf at the joint posterior mode for the alternative models considered in this document.

		-lnLikelihood						-lnPriors						Survey bias	
	$-lnPosterior$	$-lnL$	$-lnL^{Nov}$	$-lnL^{Egg}$	$-lnL^{rec}$	$-lnL^{sur}$	$-lnL^{com}$	ε_y	All growth parameters	δ_q	$N_{1983,a}$	η_y^j	k_N	k_N	k_r
A_{BH}	-795.62	-859.24	1.62	5.12	36.70	-546.01	-356.68	45.19	4.50	0.50	13.36	0.00	-0.06	0.93	0.66
A_{2BH}	-800.87	-857.51	4.00	4.99	36.32	-547.10	-355.72	38.21	4.50	0.51	13.36	0.00	-0.08	0.93	0.66
A_{3BH}	-805.91	-854.37	4.44	4.80	38.00	-545.06	-356.54	29.91	4.61	0.50	13.36	0.00	-0.06	0.94	0.67
A_{HS}	-794.82	-859.25	1.35	5.27	36.75	-545.77	-356.85	46.00	4.52	0.50	13.36	0.00	-0.09	0.92	0.66
A_{2HS}	-800.52	-856.94	5.91	4.94	36.20	-547.31	-356.68	37.92	4.62	0.50	13.36	0.00	-0.10	0.91	0.64

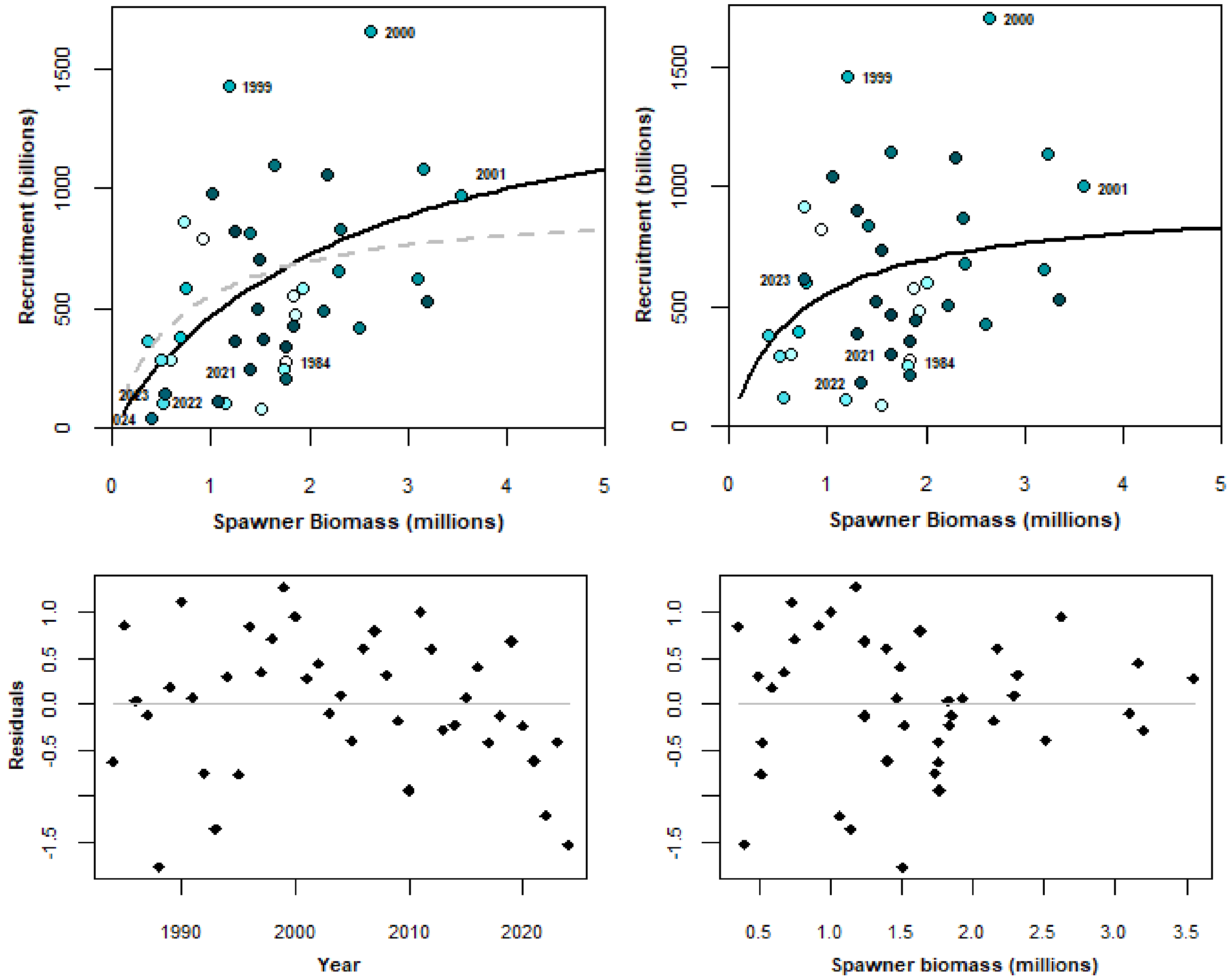

Figure 1a. Model predicted anchovy recruitment (at 1 October) plotted against spawner biomass from October 1984 to 2024 with the Beverton Holt stock recruitment relationship for A_{BH} (top left). The top right plot shows the relationship estimated by de Moor (2024) and this relationship is shown in grey in the top left plot. The residuals from the A_{BH} fit are given in the lower plots, against year and against spawner biomass. Note that the final year recruitment may not be well estimated.

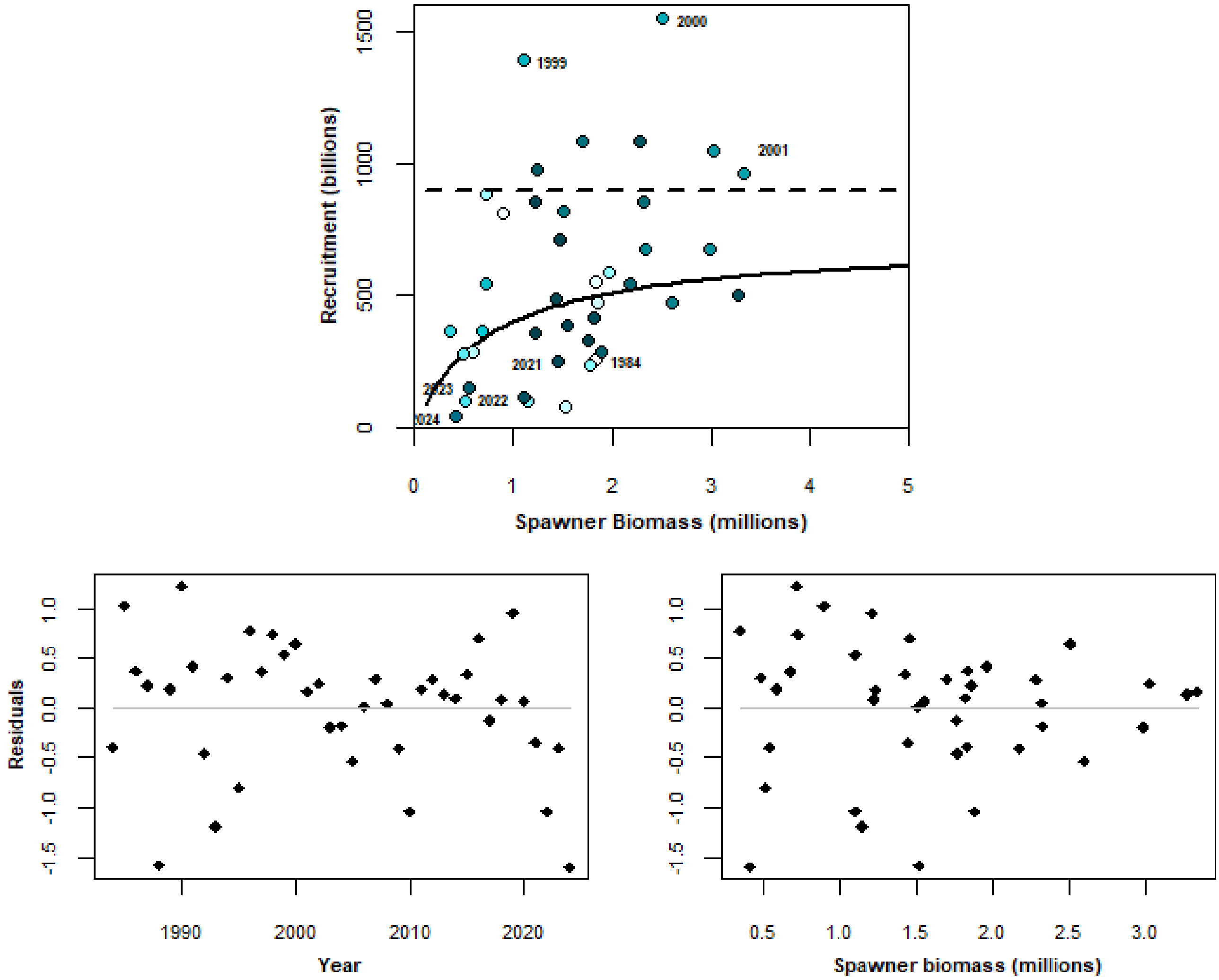

Figure 1b. Model predicted anchovy recruitment (at 1 October) plotted against spawner biomass from October 1984 to 2024 with the Beverton Holt stock recruitment relationship for A_{2BH} (top). The solid line shows the relationship estimated for early and later years while the dashed line shows the [straight line] 'relationship' estimated for the middle years. The residuals are given in the lower plots, against year and against spawner biomass. The standard deviation for the early and later years ($\sigma_r = 0.74$) is greater than that for the middle years ($\sigma_r = 0.43$). Note that the final year recruitment may not be well estimated.

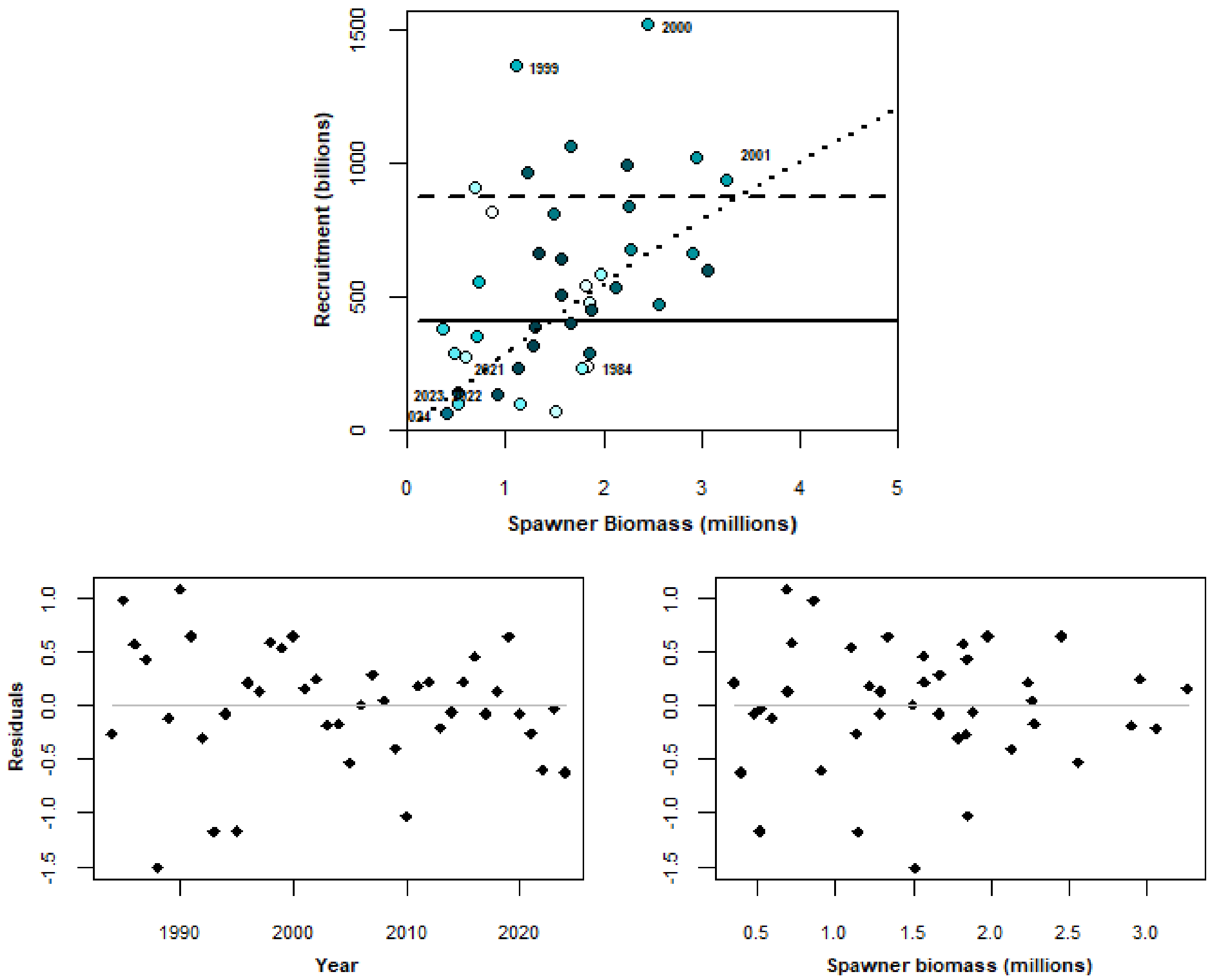

Figure 1c. Model predicted anchovy recruitment (at 1 October) plotted against spawner biomass from October 1984 to 2024 with the Beverton Holt stock recruitment relationship for A_{3BH} (top). The solid line shows the [straight line] 'relationship' estimated for early years, the dashed line shows the [straight line] 'relationship' estimated for the middle years and the dotted line shows the relationship estimated for the later years. The residuals are given in the lower plots, against year and against spawner biomass. The standard deviation for the early and later years ($\sigma_r = 0.76$) is greater than that for the middle years ($\sigma_r = 0.42$). Note that the final year recruitment may not be well estimated.

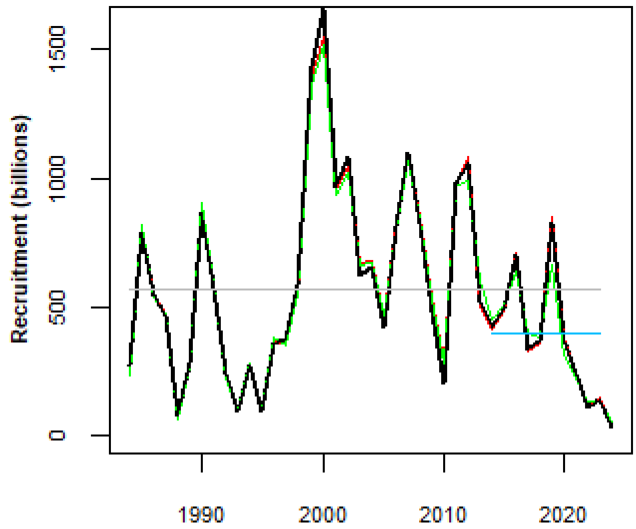

Figure 2. The timeseries of November recruitments for A_{BH} (black line), A_{2BH} (red line) and A_{3BH} (green line). The grey and blue lines indicate the A_{BH} average over the full timeseries (570 billion, decreased from 610 billion of de Moor (2024)) and the most recent 10 years (402 billion, decreased from 481 billion of de Moor (2024)), respectively, excluding November 2024.

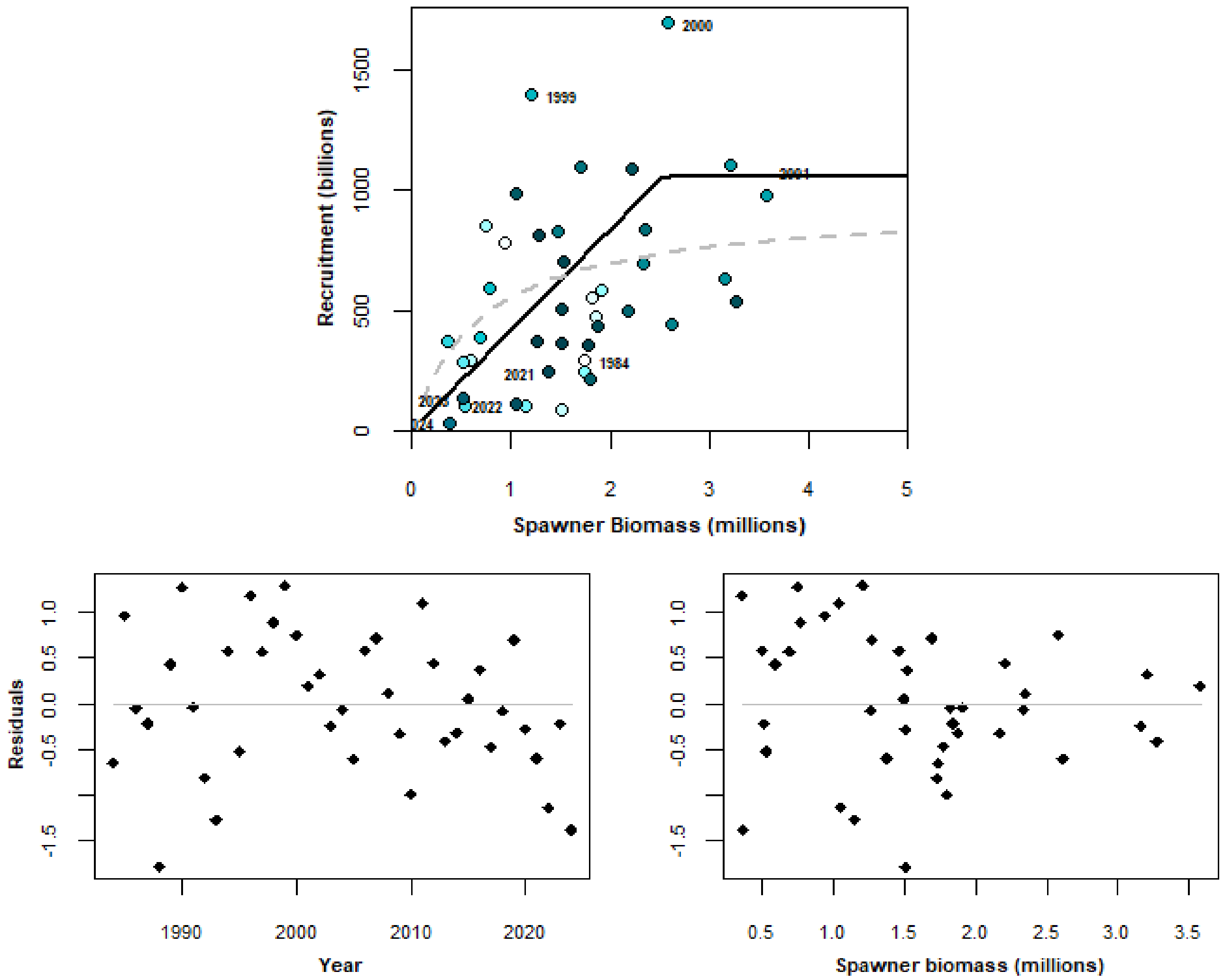

Figure 3a. Model predicted anchovy recruitment (at 1 October) plotted against spawner biomass from October 1984 to 2024 with the Hockey Stick stock recruitment relationship for A_{HS} (top). The relationship estimated by de Moor (2024) is shown in grey in

this plot. The residuals from the A_{HS} fit are given in the lower plots, against year and against spawner biomass. Note that the final year recruitment may not be well estimated.

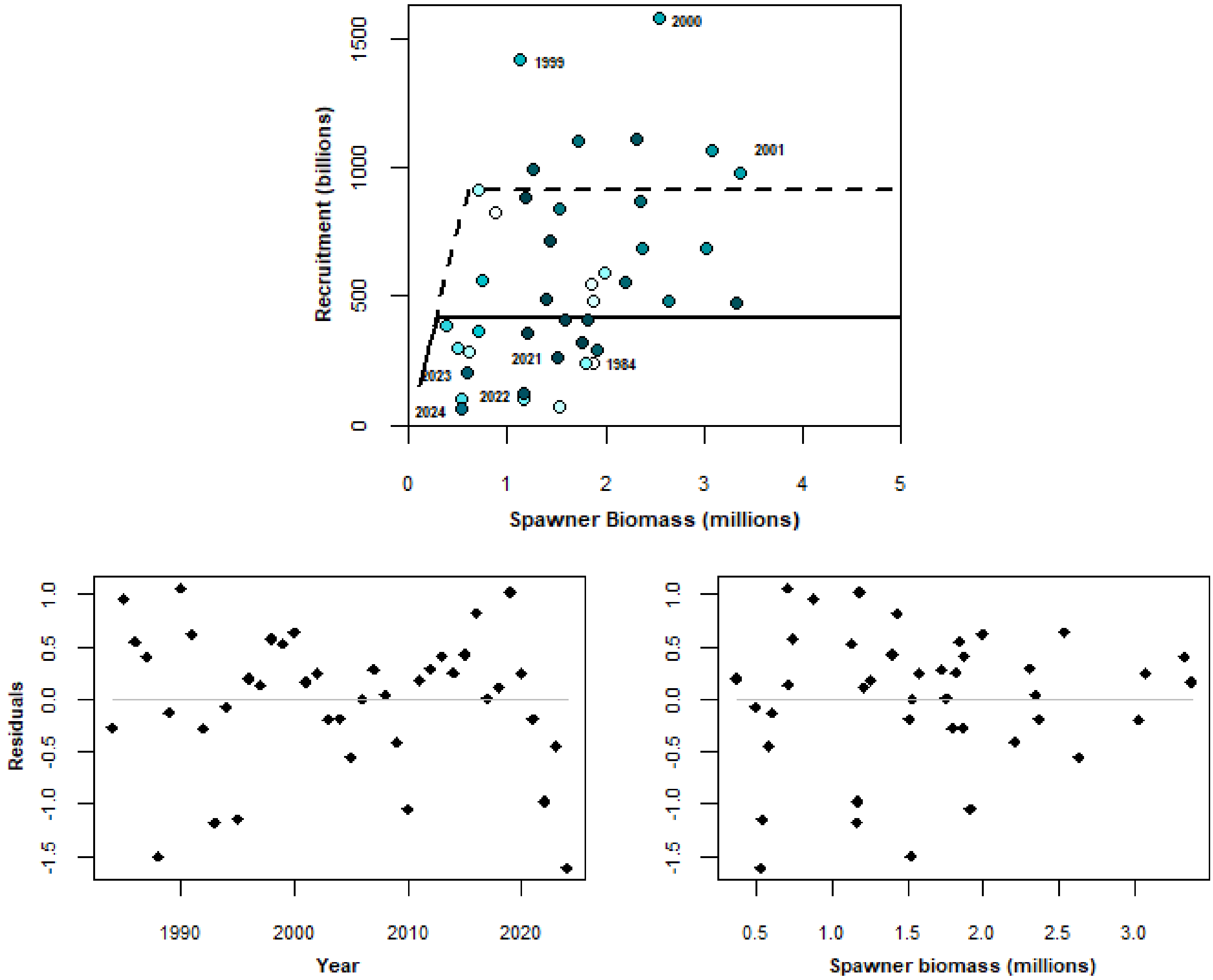

Figure 3b. Model predicted anchovy recruitment (at 1 October) plotted against spawner biomass from October 1984 to 2024 with the Hockey Stick stock recruitment relationship for A_{2HS} (top). The solid line shows the relationship estimated for early and later years while the dashed line shows the relationship estimated for the middle years. The residuals are given in the lower plots, against year and against spawner biomass. The standard deviation for the early and later years ($\sigma_r = 0.73$) is greater than that for the middle years ($\sigma_r = 0.43$). Note that the final year recruitment may not be well estimated.